Shekinah Educational Service
RC 1433060

IELTS CLASSROOM

Setup Guide

Your own online classroom on GitHub + Supabase + JaaS video, with marking by Claude

Built for	Dickson Ogaga Oweibo · 14-day intensive IELTS Academic course
Prepared by	Shekinah Educational Services · October 2026
Time needed	About 1 hour, once. Use a laptop or desktop with Google Chrome.
Cost	Free (GitHub Pages, Supabase free plan, JaaS free Dev plan)

What is in the zip file

File / folder	What it is for
1-upload-to-github	The website. You upload everything inside this folder to GitHub (Part E).
2-supabase-setup.sql	Creates the database, security rules, marking functions and all 14 days of lessons. You paste it into Supabase (Part B). **Never upload this to GitHub**: it contains the answer keys.
3-video-function-jaas-token.ts	The small program that lets you and Dickson join the video call securely. You paste it into Supabase (Part D).
Setup-Guide.pdf / .docx	This guide.

Before you start

- A laptop or desktop computer with **Google Chrome**. Setting up on a phone is very hard.
- An email address for you and one for Dickson (it can be his own email, or one you create for him).
- Open a notepad (Notepad on Windows, Notes on Mac) to paste your keys as you collect them. You will collect **five** things: Supabase Project URL, Supabase publishable (anon) key, JaaS App ID, JaaS Key ID and the JaaS private key file.

Supabase and GitHub change their screens from time to time. If a button has a slightly different name, look for the closest match. The steps stay the same.

Part A: Create three free accounts (about 15 minutes)

Step 1. GitHub (hosts the website): go to **github.com** and click **Sign up**. Enter your email, a password and a username, then verify your email with the code GitHub sends. Your username becomes part of the web address, for example `shekinah-ielts.github.io`.

Step 2. Supabase (logins, saved work, marking, recordings, live whiteboard): go to **supabase.com** and click **Start your project**. Choose **Continue with GitHub** (easiest), then allow access.

Step 3. JaaS by 8x8 (video calls): go to **jaas.8x8.vc** and sign up for the free **Dev** plan. Verify your email.

If JaaS asks for a credit card or will not accept your country, stop at this step and tell Claude. The classroom can use a Google Meet link instead, and everything else (whiteboard, chat, lessons, marking) still works.

Part B: Set up Supabase (about 20 minutes)

Step 4. In Supabase click **New project**. Fill in:

- **Name:** `ielts-classroom`
- **Database password:** click **Generate a password** and save it in your notepad. You will rarely need it.
- **Region:** choose a **Europe** region (London, Ireland or Frankfurt). These are closest to Nigeria.
- **Plan:** Free.

Step 5. Click **Create new project** and wait about 2 minutes until the project is ready.

Step 6. Create the two accounts. In the left menu click **Authentication**, then **Users**, then **Add user** → **Create new user**:

- Enter **your** email and a password, tick **Auto Confirm User**, and click **Create user**.
- Do the same for **Dickson** (his email and a password you will give him). Use at least 8 characters.

Step 7. In the left menu click **SQL Editor**, then **New query** (or the **+** button).

Step 8. Open `2-supabase-setup.sql` from the zip with Notepad (right-click → Open with → Notepad). Press **Ctrl+A** then **Ctrl+C** to copy everything. Click in the Supabase SQL box, press **Ctrl+V** to paste, then click **Run** (or press Ctrl+Enter).

> Supabase may warn that the query contains "destructive operations". That is only because it replaces old security rules. Click **Run this query** to continue. Running the file again later is safe: it never deletes Dickson's work.

Step 9. At the bottom you will see a **Setup check** table. Every line should say ✔ **OK**, except "Tutor account set", which you fix in the next step.

Step 10. Make yourself the tutor. Click **New query**, paste the lines below, **change the two email addresses** to the ones you used in Step 6, and click **Run**:

```
update public.profiles set role = 'tutor', full_name = 'Hajju'
  where lower(email) = lower('YOUR-EMAIL@example.com');
update public.profiles set full_name = 'Dickson Ogaga Oweibo'
  where lower(email) = lower('DICKSONS-EMAIL@example.com');
select email, full_name, role from public.profiles;
```

The result should list both accounts: yours with role **tutor** and Dickson's with role **student**.

Step 11. Collect your Supabase keys. Click the **Connect** button at the top of the project (or go to **Project Settings** → **API Keys** and **Data API**). Copy into your notepad:

- **Project URL**: looks like `https://abcdefghijkl.supabase.co`
- **Publishable key** (starts with `sb_publishable_`) or the legacy **anon public** key (a long text starting with `eyJ`). Either works.

> Never copy the **secret** or **service_role** key anywhere. The site does not need it.

Part C: Get your JaaS keys (about 10 minutes)

Step 12. Sign in at **jaas.8x8.vc**. On your dashboard, find your **AppID**. It starts with `vpaas-magic-cookie-`. Copy it into your notepad.

Step 13. Open **API Keys** and click **Add API key** → **Generate API key pair**. Give it a name such as `classroom`.

Step 14. **Download the private key file straight away.** JaaS shows it only once. If you miss it, delete that key and generate a new one.

Step 15. In the API Keys list, copy the new key's **Key ID** into your notepad. It may look like `vpaas-magic-cookie-…/a1b2c3` or just `a1b2c3`; both work.

Part D: Turn on secure video passes (about 10 minutes)

Step 16. Back in Supabase, click **Edge Functions** in the left menu, then **Deploy a new function** → **Via Editor**.

Step 17. Name the function exactly `jaas-token`. Delete the sample code in the editor. Open `3-video-function-jaas-token.ts` from the zip in Notepad, copy everything, paste it into the editor, and click **Deploy function**.

Step 18. Open the `jaas-token` function's **Details** (or **Settings**) and switch **Enforce JWT verification** (also called **Verify JWT**) **OFF**, then save. The function checks the sign-in itself.

Step 19. Add three secrets. Go to **Edge Functions** → **Secrets** (sometimes under **Project Settings** → **Edge Functions**). Add each name exactly as written, with its value, and save:

Name	Value
JAAS_APP_ID	Your AppID from Step 12 (vpaas-magic-cookie-…)
JAAS_KEY_ID	The Key ID from Step 15
JAAS_PRIVATE_KEY	Open the private key file from Step 14 in Notepad. Copy **everything**, including the lines `-----BEGIN … KEY-----` and `-----END … KEY-----`, and paste it as the value.

Part E: Put the website online (about 15 minutes)

Step 20. On GitHub, click the **+** at the top right → **New repository**. Name it `ielts-classroom`, choose **Public**, leave everything else as it is, and click **Create repository**.

Step 21. On the new, empty repository page, click the link **uploading an existing file**. Open the zip's `1-upload-to-github` folder on your computer, select **everything inside it** (Ctrl+A), and drag it into the GitHub page. Wait for all files to appear, including the `img` folder, then click **Commit changes**.

Step 22. Click `config.js` in the file list, then the **pencil icon** (Edit). Paste your three public values between the quotes, replacing the PASTE-… text:

```
window.CLASSROOM_CONFIG = {
  supabaseUrl: "https://abcdefghijkl.supabase.co",
  supabaseKey: "sb_publishable_…  (or the anon key)",
  jaasAppId: "vpaas-magic-cookie-…",
};
```

Click **Commit changes…** and then **Commit changes** again.

> Only these three values go in config.js. The JaaS private key and Key ID stay in Supabase secrets. Anyone can read files on a public GitHub site.

Step 23. Go to the repository's **Settings** → **Pages**. Under **Build and deployment**, set **Source** to **Deploy from a branch**, choose **main** and **/ (root)**, and click **Save**.

Step 24. Wait 1–2 minutes and refresh the Pages screen. Your link appears at the top, for example `https://yourname.github.io/ielts-classroom/`. Open it. You should see the sign-in page.

Part F: Test everything (about 10 minutes)

Step 25. On your laptop, open your link and sign in with **your** email. You should see the **Dashboard** with Dickson's name and the 14 days. Day 1 is already unlocked.

Step 26. On your phone (or in a private Chrome window), sign in as **Dickson**. You should see "Good morning, Dickson" and Day 1.

Step 27. On both devices open **Classroom**. Check that each one says the other person "is here", draw on the whiteboard (it appears on the other screen), send a chat message, and click **Join video call** on both.

Step 28. As Dickson, open Day 1 → Exercises → the Reading check, answer a couple of questions and submit. As tutor, open Day 1 on your dashboard (**Mark**) and check it appears, already auto-marked.

Step 29. Clean up the test: on the tutor marking page click **Reopen for student** on that exercise, then use **Settings** → **Clear chat** and **Clear whiteboard**.

Step 30. Send Dickson his link, email and password. Here is a short message you can use:

> Hi Dickson, your IELTS classroom is ready: [your link]. Sign in with [his email] and the password [password]. Use Google Chrome on a laptop if you can, and allow the microphone when asked. Day 1 is open: start with the lesson, then the exercises. See you in class!

Your daily routine

When	What you do
Before class	On the **Dashboard**, switch on the day (if it is not already unlocked).
In class	Open **Classroom** → **Join video call**. Use the **Lesson** tab to present (Dickson's screen follows yours). Use the **Whiteboard** to explain, and **Push to board** to put a chart, passage or his essay on a new page. Start the **timer** for timed tasks.
Exercises	Dickson does them on his own device (in class or as homework). Listening plays once, like the real test. Writing has a word count and timer. Speaking is recorded with a transcript.
After he submits	Open the day's **Mark** page: **1 Export for Claude** → **2 Import marking** → check it → **3 Release to student**.
Next class	Go through the feedback together. He writes his **Corrections**; you review them on the Mark page, then **4 Unlock** the next day.
Mock days (6, 7, 10, 12, 14)	He uses a Cambridge IELTS Academic book and types his 40 answers into the answer sheet. On the Mark page, paste the book's answer key (one per line) and click **Save key**. It marks itself and estimates the band.

Marking with Claude (free, no API key)

Step 1. On the day's Mark page click **Export for Claude**. A file downloads, for example `dickson-ogaga-oweibo-day3-for-claude.json`.

Step 2. Open a Claude chat (claude.ai, or this Shekinah project), attach the file and type: "Mark this IELTS work and give me the marking file." The file contains all the instructions Claude needs.

Step 3. Download the file Claude gives back (`marking-…-day3.json`). If Claude replies with text instead of a file, copy the whole reply.

Step 4. Back on the Mark page click **Import marking** and choose the file (or paste the text). Every exercise fills in at once.

Step 5. Check the marking. Change anything you disagree with, add the **Pronunciation** score for speaking tasks (Claude can't hear the audio), then click **Release to student**.

> Listening and Reading are already marked against the answer key when he submits. Claude explains each wrong answer and can accept a valid alternative spelling. You can click any ✓/✗ yourself to override it.

Troubleshooting

What you see	What to do
"Almost there": the site isn't connected	config.js still has PASTE-… text, or a value is wrong. Edit it on GitHub (Step 22), wait a minute, then press **Ctrl+F5**.

What you see	What to do
“Wrong email or password”	Check the account exists in Supabase → Authentication → Users. As tutor you can set Dickson a new password in **Settings**. If that fails, in Supabase open the user and use the reset option, or delete and recreate the user, then run setup.sql again.
“Your account exists but is not set up yet”	Run `2-supabase-setup.sql` again (Step 8). It links any new accounts.
You see the student screens instead of the tutor dashboard	Run the make-tutor lines again (Step 10) with your exact email, then sign out and in.
A ⚠️ line in the setup check	Run the file once more. If “Recordings storage” or “Live updates” still shows ⚠️, the site still works; tell Claude the exact line and the message in the Results tab.
Video: “jaas-token Edge Function was not found”	The function name must be exactly `jaas-token` (Step 17).
Video: “Video is not set up yet”	One of the three secrets is missing or misspelt (Step 19).
Video: “Please sign in again” / 401	Turn **Enforce JWT verification** off (Step 18), then sign out and in.
Video: “Could not create the video pass … private key”	Paste the whole private key file again, including the BEGIN and END lines.
Video opens but says you are not allowed to join	The App ID in config.js and JAAS_APP_ID must match, and the Key ID must belong to the private key you pasted.
Camera or microphone blocked	Click the padlock next to the web address → allow Camera and Microphone → refresh.
Listening doesn’t play	Use Google Chrome and turn the volume up. If he already pressed Start, open the Mark page and click **Allow another play**.
Speaking has no transcript	Transcripts need Google Chrome (laptop is best). The recording still works: listen to it, and type a short transcript on the Mark page if you want Claude to mark it.
Whiteboard or chat doesn’t update live	Refresh both screens. In Supabase → **Realtime** → **Settings**, make sure public channels are allowed.
Supabase says the project is paused	Free projects pause after 7 days without use. Open the project and click **Restore**. Nothing is lost.

Keeping things safe

- **Public:** the website files on GitHub and the three values in config.js. These are designed to be public.
- **Private:** Dickson’s work, recordings, answer keys, marking and your notes are protected by the Supabase security rules. He only sees marking after you release it, and only days you unlock.
- **Keep secret:** your JaaS private key (only in Supabase secrets) and your Supabase database password. Don’t upload `2-supabase-setup.sql` to GitHub: it contains every answer key.

Later

- **Another student:** add the user in Supabase → Authentication → Users. They appear on your dashboard automatically with Day 1 unlocked.
- **Updates from Claude:** a new setup.sql is safe to run again and keeps all work. New website files replace the old ones on GitHub (upload and commit). Keep your edited config.js.